$$ds^2_{minkw} = -dt^2 + dr^2 + r^2 d\Omega^2$$

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + e^{2\gamma(r)} r^2 d\Omega^2$$

$e^{2\alpha}, e^{2\beta}, e^{2\gamma}$ are arbitrary why exponential?

because of simplicity

$\bar{r} = e^{\gamma(r)} r$ let us define $\bar{r}$

$$\bar{r} = e^{\gamma(r)} r \Rightarrow d\bar{r} = e^{\gamma} r\, d\gamma + e^{\gamma} dr$$

$$d\bar{r} = e^{\gamma} dr\left(r \frac{d\gamma}{dr} + 1\right) \Rightarrow dr = \frac{d\bar{r}}{\left(r\frac{d\gamma}{dr}+1\right)} \cdot e^{-\gamma}$$

$$dr^2 = \frac{d\bar{r}^2}{\left(r\frac{d\gamma}{dr}+1\right)^2} \cdot e^{-2\gamma}$$

$$\Rightarrow ds^2 = -e^{+2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + e^{2\gamma(r)} r^2 d\Omega^2 \qquad r = \bar{r} e^{-\gamma(r)}$$

$$\Rightarrow ds^2 = -e^{+2\alpha} dt^2 + d\bar{r}^2 \cdot \left(r\frac{d\gamma}{dr}+1\right)^{-2} \cdot e^{2\beta - 2\gamma} + e^{2\gamma} r^2 d\Omega^2$$

$$ds^2 = -e^{+2\alpha} dt^2 + d\bar{r}^2 \cdot \left(r\frac{d\gamma}{dr}+1\right)^{-2} e^{2\beta-2\gamma} + \bar{r}^2 d\Omega^2$$

$$\bar{r} \to r \qquad \left(1 + r\frac{d\gamma}{dr}\right)^{-2} e^{2\beta - 2\gamma} \to e^{2\beta'(r)}$$

We are able to write the metric, in a simpler way

$$ds^2 = -e^{2\alpha} dt^2 + dr^2 e^{2B(r)} + r^2 d\Omega^2$$

we need to derive Christoffel Symbols to know Curvature that is required for Einstein Field Equation

$$\Gamma^{\rho}_{\mu\nu} = \frac{1}{2} g^{\rho\lambda} [\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}]$$

we can solve

$$\Gamma^{t}_{tr} = \partial_r \alpha \qquad \Gamma^{r}_{tt} = e^{2(\alpha - B)} \partial_r \alpha \qquad \Gamma^{r}_{rr} = \partial_r B$$

$$\Gamma^{\theta}_{r\theta} = \frac{1}{r} \qquad \Gamma^{r}_{\theta\theta} = -r e^{-2B} \qquad \Gamma^{\phi}_{r\phi} = \frac{1}{r}$$

$$\Gamma^{r}_{\phi\phi} = -r e^{-2B} \sin^2\theta \qquad \Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta \qquad \Gamma^{\phi}_{\theta\phi} = \frac{\cos\theta}{\sin\theta}$$

$$\Gamma^{\rho}_{\mu\nu} = \Gamma^{\rho}_{\nu\mu} \quad \text{(Non Torsion) symmetric}$$

$$R^{i}_{jkl} = \frac{\partial \Gamma^{i}_{lj}}{\partial u_k} - \frac{\partial \Gamma^{i}_{kj}}{\partial u_l} + \sum_{m=1}^{n} (\Gamma^{i}_{km} \Gamma^{m}_{lj} - \Gamma^{m}_{kj} \Gamma^{i}_{lm})$$

$$R^{t}_{rtr} = \partial_r \alpha \, \partial_r B - \partial_r^2 \alpha - (\partial_r \alpha)^2$$

$$R^{t}_{\theta t\theta} = -r e^{-2B} \partial_r \alpha$$

$$R^{t}_{\phi t\phi} = -r e^{-2B} \sin^2\theta \, \partial_r \alpha$$

$$R^r{}_{\theta r \theta} = r e^{-2B} \partial_r B$$

$$R^r{}_{\phi r \phi} = r e^{-2B} \sin^2\theta \, \partial_r B$$

$$R^\theta{}_{\phi \theta \phi} = (1 - e^{-2B}) . \sin^2\theta$$

$R^i{}_{jkl} \Rightarrow$ What does Ricci Tensor?

$$R_{kl} = g_{i\mu} g^{\mu j} R^i{}_{jkl} \Rightarrow g_{i\mu} g^{\mu j} R^i{}_{jkl} = R_{kl}$$

$$R_{tt} = e^{2(\alpha - B)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r B + \frac{2}{r} \partial_r \alpha \right]$$

$$R_{rr} = -\partial_r^2 \alpha - (\partial_r \alpha)^2 + \partial_r \alpha \partial_r B + \frac{2}{r} \partial_r B$$

$$R_{\theta\theta} = e^{-2B} [r (\partial_r B - \partial_r \alpha) - 1] + 1$$

$$R_{\phi\phi} = \sin^2\theta \, R_{\theta\theta}$$

what does Ricci Scalar?

$$R \Rightarrow g^{kl} R_{kl} = R$$

$$R = -2e^{-2B} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - \partial_r \alpha \partial_r B + \frac{2}{r}(\partial_r \alpha - \partial_r B) + \frac{1}{r^2}(1 - e^{2B}) \right]$$

Let us consider exterior region of the Black Hole where Ricci Tensor should be zero because Energy Momentum Tensor is zero in vacuum but Riemannian Tensor is not zero exterior region of Black Hole.

$$R_{tt} = e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - \partial_r\alpha\,\partial_r\beta + \frac{2}{r}\partial_r\alpha\right]$$

$$R_{rr} = -\partial_r^2\alpha - (\partial_r\alpha)^2 + \partial_r\alpha\,\partial_r\beta + \frac{2}{r}\partial_r\beta$$

$$e^{-2(\alpha-\beta)} \cdot R_{tt} \Rightarrow$$

$$R_{rr} + e^{-2(\alpha-\beta)}R_{tt} = 0$$

$$\left[\cancel{\partial_r^2\alpha} + \cancel{(\partial_r\alpha)^2} - \cancel{\partial_r\alpha\,\partial_r\beta} + \frac{2}{r}\partial_r\alpha\right]$$

$$\cancel{-\partial_r^2\alpha} - \cancel{(\partial_r\alpha)^2} + \cancel{\partial_r\alpha\,\partial_r\beta} + \frac{2}{r}\partial_r\beta$$

$$\Rightarrow R_{rr} + e^{-2(\alpha-\beta)}R_{tt} = 0 = \frac{2}{r}\partial_r(\alpha+\beta)$$

if we take derivative and conclusion is a zero

$$\partial_r(\alpha+\beta) = 0 \Rightarrow \alpha = -\beta + c \qquad c \text{ is constant}$$

$\Rightarrow t \to e^{-c} t \Rightarrow$ we can take c is zero

$\Rightarrow \alpha = -\beta$ Now let us consider

$R_{\theta\theta} = e^{-2\beta}[r(\partial_r \beta - \partial_r \alpha) - 1] + 1$

$R_{\phi\phi} = \sin^2\theta \, R_{\theta\theta}$

If $R_{\theta\theta}$ is zero, $R_{\phi\phi}$ will be zero

$\alpha = -\beta \Rightarrow R_{\theta\theta} = e^{2\alpha}[r(-\partial_r \alpha - \partial_r \alpha) - 1] + 1$

$\Rightarrow R_{\theta\theta} = -e^{2\alpha}[2r(\partial_r \alpha) + 1] + 1 = 0$

$e^{2\alpha}(2r(\partial_r \alpha) + 1) = +1$

$\frac{d}{dr}(r e^{2\alpha}) = e^{2\alpha} + 2\frac{d\alpha}{dr} \cdot r e^{2\alpha} = e^{2\alpha}\left(1 + 2r\frac{d\alpha}{dr}\right)$

$\Rightarrow \frac{d}{dr}(r e^{2\alpha}) = 1 \Rightarrow$ which function take derivative $1 = ?$

$\Rightarrow e^{2\alpha} = 1 - \frac{R_s}{r} \Rightarrow \frac{d}{dr}\left(1 - \frac{R_s}{r}\right) = +r^{-2}R_s$

$\frac{d}{dr}(r e^{2\alpha}) = 1 \Rightarrow e^{2\alpha} + r\frac{d}{dr}\left(1 - \frac{R_s}{r}\right) \Rightarrow e^{2\alpha} + r \cdot r^{-2}R_s$

$$\Rightarrow \quad e^{2\alpha} + r^{-1} R_S = 1 - \cancel{\frac{R_S}{r}} + \cancel{\frac{1}{r} R_S} = 1 \quad \text{correct} \checkmark$$

$$ds^2 = -e^{2\alpha} dt^2 + dr^2 e^{2B(r)} + r^2 d\Omega^2 \qquad B' = -\alpha$$

$$\Rightarrow \quad ds^2 = -\left(1 - \frac{R_S}{r}\right) dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Schwarzchild metric is proved.

R_S refers to the radius of Schwarzchild and it determines the horizon.